Social Isolation in Adolescence: Its Effects on Relationship Building and Individual Function

Jingxuan Nie

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Abstract. Social interaction is important during adolescence, as it affects how young people build relationships, regulate emotions, and function in daily life. This paper reviews peer-reviewed psychology and education research to ask how isolation affects those abilities and how the learning environment changes the effect. A lack of interaction may produce negative impacts on mental health, identity formation, and brain development, and poor social relationships are associated with a 29 percent increased risk of heart disease and a 32 percent increased risk of stroke, while around 75 percent of mental disorders begin before age 24. Isolation is divided into involuntary and voluntary forms. Involuntary isolation is driven by misinterpretation of others' willingness to socialize and by the absence of a relationship strong enough to support self-disclosure, producing a cycle in which rejection reduces future disclosure. Voluntary solitude is associated with feeling less stressed and more self-connected, yet higher unsociability still correlates with peer exclusion, depression, and lower self-esteem. On environment, online learning sustains more emotional engagement than is commonly assumed and can support relationships where teachers facilitate interaction, but in-person learning remains stronger for teamwork and concentration. The paper concludes that in-person institutions should provide more time for students to interact and more extracurricular range. The findings are associational and several sources study college students rather than adolescents.

Keywords. adolescence; social isolation; solitude; learning environments; relationship formation

1. INTRODUCTION

Social interaction is extremely important during one's adolescence, as it affects how they build their relationships, regulate their emotions, and function in daily life. During this period adolescents are prone to social development, meaning that a lack of interaction may result in several negative consequences, including negative impacts on the adolescent's mental health, formation of identity, and development of their brain, according to Amy Orben and colleagues, writing in The Lancet Child & Adolescent Health (Orben et al., 2020). This shows the importance of developing social skills early on, as it sets up one's ability to establish connections in the future. Orben also notes that poor relationships, including those affected by peer rejection, bullying, and loneliness, are linked to many different mental health risks.

In addition, Vivek H. Murthy, the Surgeon General from 2021 to 2025, reports that poor social relationships are associated with a 29 percent increased risk of heart disease and a 32 percent increased risk of stroke (Murthy, 2023). This is supported by Sundqvist et al. (2024), writing for the International Journal of Qualitative Studies on Health and Well-being, who note that the onset of most mental disorders occurs during childhood, adolescence, or the early adult years, with around 75 percent starting before the age of 24. Overall, socialization during the early stages of life is extremely important, as it establishes a base for one's skills in connecting with others and lessens the risk of negative effects on mental and physical health.

While voluntary isolation can help adolescents manage stress and allow for self-reflection, a lack of meaningful interaction harms their ability to establish relationships and impacts their ability to function. Adolescents who experience loneliness or stress may become less confident in social situations, and may lose focus, self-regulation, and the ability to work while isolated.

2. TWO KINDS OF ISOLATION

To better understand this issue, isolation should not be limited to one single definition. Rather, it should be divided into involuntary and voluntary isolation. Part of this idea is found in Chen et al. (2024), writing for Frontiers in Psychiatry.

Involuntary isolation. This is defined as separation from others against one's will, the classic idea that adolescents are not able to socialize for varying reasons. Melissa de Witte, writing for Stanford, cites Stanford psychologist Jamil Zaki, who notes that adolescent isolation may be caused by misinterpretation of others' desire to socialize (De Witte, 2025). De Witte views this as a cycle, where adolescents try to socialize with others, get rejected because others are not open to socializing at the time, and then become more isolated due to hesitation about trying to establish connections again.

Another reason adolescents may not be able to socialize is that they have no base to begin from. Costello et al. (2024), researchers in psychology at the University of Virginia writing for the Journal of Research on Adolescence, establish the idea of a virtuous cycle, in which relationships high in closeness allow for self-disclosure, and self-disclosure fosters intimacy, so that both increase over time. The idea behind this is that if one has a relationship open to disclosure and inclusivity, it results in heightened relationship quality. However, if an adolescent does not have a strong relationship to begin with, it may be difficult to strengthen that relationship through disclosure due to fear of rejection. Costello furthers this, stating that teens whose disclosure is met with rejection may not disclose again, reducing learning opportunities with that friend and with other friends in the future.

Voluntary isolation. This is defined as an adolescent's desire to be alone and separated from others. Chen aids this definition by stating that during the transition from childhood to adolescence, individuals' preference for solitude increases as they grow older, which shows that a large part of isolation found in adolescent years is voluntary and desired. In addition, Netta Weinstein et al. (2023), writing for Scientific Reports, add that solitude helps with feeling less stressed and more peaceful and relaxed, and may provide other benefits including fostering self-connection and a sense of autonomy.

However, voluntary isolation still has the downsides of being isolated. Chen furthers the potential issues by noting that higher unsociability is related to higher peer exclusion and internalizing problems, and to lower prosocial behaviours and interpersonal skills, with these associations buffered by children's resilience, and that higher unsociability is related to higher depression and loneliness and lower self-esteem. This displays that while it is the adolescent's decision to be alone, and they may feel calmer and more autonomous, they still face the harms of remaining isolated. The choice changes the cause, not the effect.

Altogether, a major reason adolescents find it difficult to socialize is that they may have a weak foundation or none at all in terms of relationships. Both Zaki and Costello address this, with Zaki noting that adolescent misinterpretation of others may result in a cycle of isolation, and Costello discussing how teens faced with rejection after self-disclosure may end up in a weaker relationship.

3. THE LEARNING ENVIRONMENT

While there are a variety of sociological factors in an adolescent's isolation, one's physical environment may also have a large influence. One of the largest factors affecting student learning and social connectivity is the learning environment, because learning institutions tend to promote social connectivity, and depending on the type of institution the adolescent attends, it may have a range of effects on their sociability.

3.1 Online learning

To begin, learning in an online environment is typically viewed negatively. A common generalization is that adolescents do not experience the full range of emotions and social interactions online that they would in person, such as feeling happy and connected. However, Katrina A. Meyer and Stephanie J. Jones (2012), researchers in education, argue against this, noting that students still displayed emotions typically seen in an in-person environment. After surveying 67 students from Texas Tech University and the University of Memphis, Meyer and Jones found that the two emotions with the highest means were interest and surprise, and the lowest mean captured anger, which shows that in an online learning environment students can still experience positive emotions and are not met with a purely study-focused environment.

In addition, the research found that students tend to emit more positive emotions when reading an email sent by a friend or colleague. Meyer and Jones found that when reading these emails, 46.4 percent of students reported laughing out loud frequently, 49.1 percent chuckled frequently, and 50.9 percent smiled frequently, with further students reporting that they did so all of the time. These are frequency ratings rather than counts of students, and they show that through online interactions students can still experience interconnectivity as they would in person.

Similarly, Gao et al. (2024), writing for Behavioral Sciences, find that online learning can help establish relationships between students and teachers as long as interactions are facilitated by the teacher. After conducting a study of 344 students, Gao states that social presence and learning engagement serially mediated the relationship between learner-instructor interactions and learning efficiency, which shows that with facilitation by the teacher, students can establish relationships online.

3.2 In-person learning

While online learning can establish an environment where students connect and interact with others, it is still not as effective as in-person learning in terms of distractions and teamwork (Photopoulos et al., 2022). Learning in a home environment can have a negative impact on one's learning abilities, as low motivation, low concentration, and distraction caused by the home environment are commonly reported problems during emergency remote teaching, according to Photopoulos et al. The effects of learning in a home environment resulted in impeded learning and engagement.

In addition, teamwork is more effective when done face to face, with students believing that non-face-to-face teaching does not benefit communication between classmates. While the article agrees that online learning results in less time wasted commuting to campus and provides easy accessibility, students still tend to express a desire to return to campus in order to maintain relationships with others. To further this, Photopoulos states that the classroom environment offered a higher degree of positive feelings, and that the loss of interaction and communication with instructors and fellow students has been described as a source of frustration and diminished learning.

In addition, Ferreira et al. (2021), writing for Geoforum, an interdisciplinary journal on human geography, further this by discussing how coffee shops, public places that support the establishment of connections, allow people to spend time to relax, socialize, and visit for the joys of association. They describe the third place as somewhere in which people from a diversity of backgrounds combine to expand one another's understanding of the world, and out of the bonds formed there, community takes root. This shows the importance of schools trying to establish an environment where students can connect and bond with one another.

3.3 Homeschooling

Alternatively, rather than learning online or in person, research conducted by McKinley et al., writing for the Department of Psychology at Mount Saint Mary's University, found that students who are homeschooled have been found to be more social than those in online or in-person learning environments. McKinley states that home-schooled children are often involved in many extracurricular activities that provide social contact, which shows that adolescents who are homeschooled may experience more interactions with others and therefore are more likely to establish friendships and lessen isolation, whereas in online and in-person environments students have limited time interacting with others, as they are primarily focused on learning and only given set times to interact.

In addition, McKinley finds that home-schooled children are more likely to be exposed to a wide variety of people, rather than being limited to contact with same-age, demographically similar children as children in public schools are, which displays that homeschooling allows for a wider range of interactions and helps adolescents establish a wider range of social connections. This finding runs against the common assumption and rests on a single study, so it is reported here as the source states it rather than as settled.

4. DISCUSSION

All in all, while there are many upsides to online learning, such as effectiveness and easier access (Photopoulos et al., 2022), and to homeschooling, such as more social interactions with a wider range of people (McKinley et al.), in-person learning presents itself as the best environment for students to establish connections and lessen isolation, as in-person interaction establishes stronger relationships and promotes teamwork.

The more useful point is that the advantage is not automatic. Gao et al. (2024) show that online environments support relationships when a teacher facilitates interaction, and Photopoulos et al. show that in-person settings lose their advantage when the time for interaction is missing. What matters in both cases is whether interaction is deliberately created. In-person learning institutions should therefore provide more time for adolescent students to interact with others in order to deepen their relationships, as this provides a time of disclosure (Costello et al., 2024), and should promote more time spent on extracurricular activities in order to broaden the range of people adolescents interact with (McKinley et al.).

5. LIMITATIONS

This report has several limitations.

Population mismatch. Some of the studies used in this paper focus on college students or adults rather than specifically on adolescents, which may result in the findings not fully relating to younger students and teens.

Geographic scope. Some studies focus on adolescents and students outside of the United States, which may not be applicable to students in America due to differences in culture and socioeconomic status.

Limited research base. Research conducted on isolated students and teens in the United States is relatively limited, making it harder to draw conclusions. The homeschooling comparison in particular rests on one source.

No causal claims. The relationships described here are associational. None of the evidence reviewed establishes that isolation causes the outcomes with which it correlates.

Although these limitations reduce the generalizability of the findings, the research has strongly indicated that reduced interaction and isolation have a negative effect on adolescent mental health and on the ability to establish relationships in the future. Physical learning environments appear to be the most effective in terms of reducing isolation and establishing connections, as long as schools openly promote connectivity and provide time for students to establish connections.

6. CONCLUSION

In conclusion, a lack of social interaction with others generally equates to a worsened ability to establish relationships and negative impacts on mental health. While voluntary isolation may be beneficial, involuntary isolation is much more damaging, as it increases loneliness, discourages trust, and makes it more difficult to establish new connections. The effects of isolation can also be shaped by one's learning environment, as students can establish more connections in a more socially supportive environment. Forming bonds at learning environments can result in stronger relationships, as adolescents stay engaged.

Overall, social interaction during adolescence is extremely important for developing one's social and academic abilities and for improving mental health. A stronger follow-up study would track the same adolescents across learning environments, measuring disclosure behaviour and relationship quality directly rather than inferring them from setting.

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CONTRIBUTIONS

Jingxuan Nie, sole author, developed the research question, gathered and reviewed the sources, built the distinction between voluntary and involuntary isolation and its link to the learning environment, and wrote the manuscript.